Absolutely. I'll turn the uploaded material into **interview-focused notes**, not just a textbook summary.

The source is **Chapter 19: Random Variables and Distributions**, and it explicitly moves from random variables → distributions → PMF/CDF → important distributions → total probability → sums/convolution → continuous distributions/density.

I'll use an **80/20 approach**:

- **P0 — Must know for interviews:** definitions, intuition, formulas, diagrams, examples, implementation
- **P1 — Important:** useful follow-up concepts and common interview traps
- **P2 — Advanced:** understand conceptually, but don't spend disproportionate time memorizing

Interview Notes: Random Variables & Distributions

1. Big Picture

Before learning individual distributions, understand this flow:

Real-world uncertain experiment

|

▼

Probability Space

(Ω, Σ, P)

|

▼

Random Variable X

|

▼

Possible observations

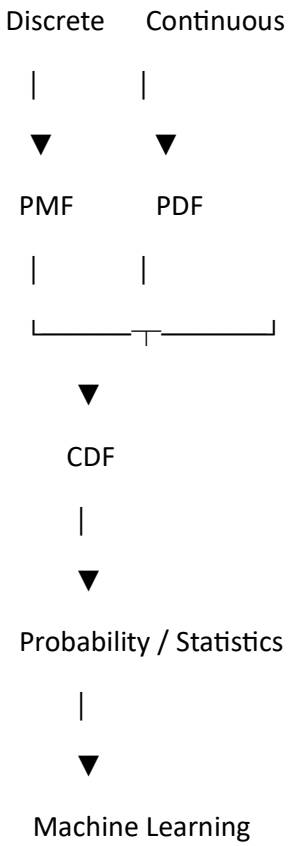
|

▼

Distribution

/ \

/ \



The key idea from the source is that a random variable is a **function that maps outcomes to measurements**. This lets us reason about quantities rather than manipulating the entire underlying probability space.

---

2. Random Variable

P0

Definition

A **random variable** is a function that maps an outcome of a random experiment to a measurable value.

For example:

Coin toss

Ω = {H, T}

Random Variable X

H ----------> 1

T ----------> 0

So:

$$X = \begin{cases} 1 & \text{if Head} \\ 0 & \text{if Tail} \end{cases}$$

The important point:

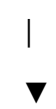
A random variable is a **function**, not necessarily a "variable that randomly changes."

The source explicitly defines random variables as functions from the probability space to a measurement space.

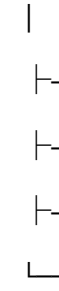
ML example

Suppose we have customers:

Customer



Random Variable X



For example:

$$X(\text{customer}) = \text{customer's purchase amount}$$

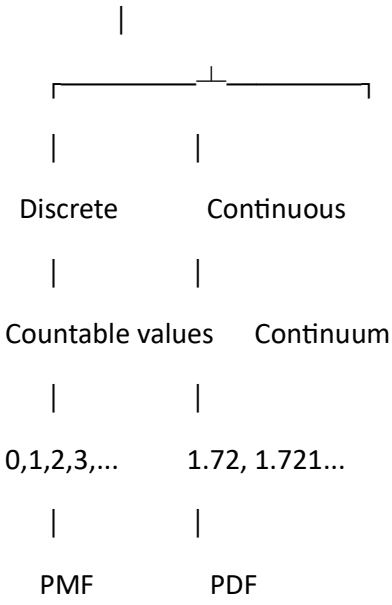
This is why random variables are fundamental to ML: **our dataset contains observations of random variables.**

---

3. Discrete vs Continuous Random Variable

P0

Random Variable



Discrete

Possible values can be counted.

Examples:

- Number of customers
- Number of heads
- Number of defects
- Number of clicks
- Number of purchases

Continuous

Possible values lie over an interval.

Examples:

- Height
- Weight
- Temperature
- Response time
- Blood pressure
- Model inference latency

The source separates discrete random variables from real-valued random variables and later develops the continuous case.

---

4. Probability Distribution

P0

A **probability distribution** tells us how probability is allocated across the possible values of a random variable.

Think:

Random Variable X

|

▼

"How frequently/probably does each value occur?"

|

▼

Probability Distribution

For a discrete variable:

X 0 1 2 3

| | | |

P(X) .10 .30 .40 .20

The probabilities must satisfy:

$$P(X = x_i) \geq 0$$

and

$$\sum_i P\,(X = x_i) = 1$$

The source explicitly gives non-negativity and total probability of 1 as the requirements for a discrete distribution.

---

5. PMF — Probability Mass Function

P0

Definition

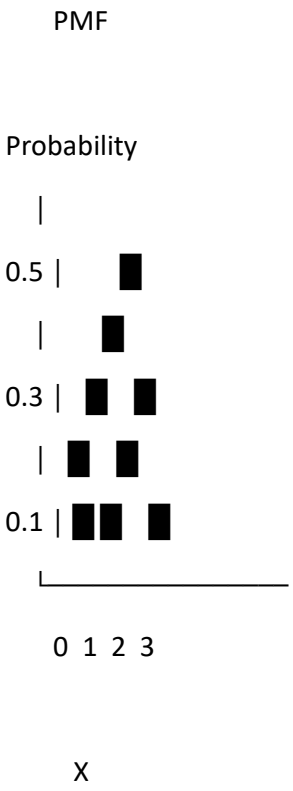
For a discrete random variable:

$$p_X(x) = P(X = x)$$

This is the **Probability Mass Function**.

The source gives exactly this definition.

Diagram



Example

Suppose:

$$P(X = 0) = 0.2$$
$$P(X = 1) = 0.5$$
$$P(X = 2) = 0.3$$

Then:

$$P(X = 1) = 0.5$$

Python

```
from scipy.stats import bernoulli


p = 0.7


print(bernoulli.pmf(1, p))
print(bernoulli.pmf(0, p))
```

The source itself demonstrates SciPy's pmf() implementation.

Interview question

Q: What is the difference between a random variable and PMF?

Answer:

A random variable maps outcomes to values. A PMF describes the probability of each possible value for a discrete random variable.

---

6. CDF — Cumulative Distribution Function

P0

Definition

$$F_X(x) = P(X \leq x)$$

The source defines CDF exactly this way.

Intuition

PMF asks:

"What is the probability of exactly this value?"

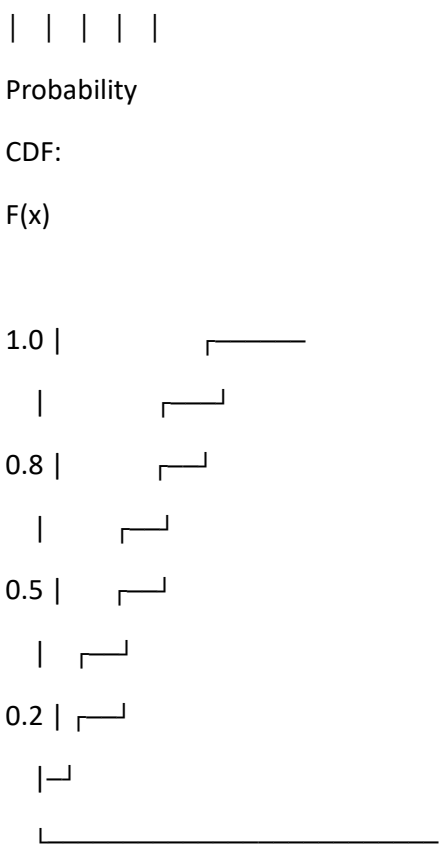
CDF asks:

"What is the probability of being at or below this value?"

Example:

Values:

```
1  2  3  4  5
```



For example:

$$F(3) = P(X \leq 3)$$

Very important property

A CDF is:

1. non-decreasing
2. right-continuous
3. approaches 0 as $x \to -\infty$
4. approaches 1 as $x \to \infty$

These properties are explicitly stated in the source.

Probability using CDF

One of the most useful formulas:

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

The source gives this relationship directly.

Interview distinction

PMF	CDF
$P(X = x)$	$P(X \leq x)$
Mainly discrete	Discrete + continuous
Exact value	Cumulative probability
Bar-like representation	Step/smooth curve

---

7. Bernoulli Distribution

P0 — VERY IMPORTANT

Bernoulli models **one binary experiment**.

Examples:

- customer churn: yes/no
- transaction successful: yes/no
- email clicked: yes/no
- fraud: yes/no
- model prediction correct: yes/no

The source defines:

$$X = \begin{cases} 1 & success \\ 0 & failure \end{cases}$$

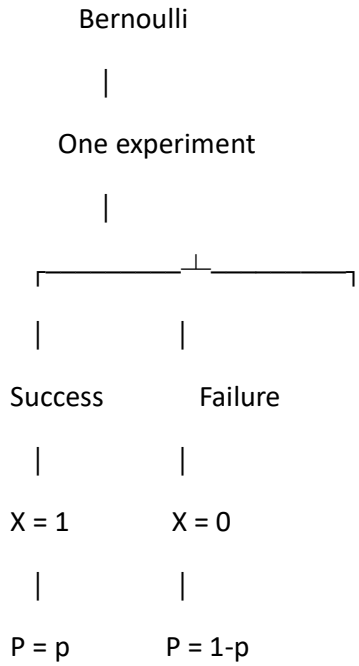
with:

$$X \sim Bernoulli(p)$$

and:

$$P(X = 1) = p$$
$$P(X = 0) = 1 - p$$

Diagram



Example

A fraud detector classifies a transaction:

fraud = 1

not fraud = 0

If:

$$P(fraud) = 0.02$$

then:

$$X \sim Bernoulli(0.02)$$

Python

```
from scipy.stats import bernoulli


p = 0.02


prob_fraud = bernoulli.pmf(1, p)
prob_not_fraud = bernoulli.pmf(0, p)
```

```
print(prob_fraud)

print(prob_not_fraud)
```

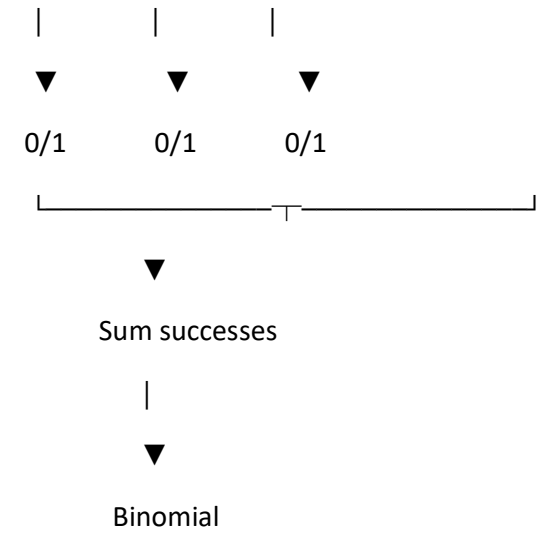
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8. Binomial Distribution

P0 — VERY IMPORTANT

Bernoulli = **one trial**

Binomial = **number of successes in n independent trials**



If:

$$X \sim Binomial(n, p)$$

then:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

The source derives the binomial distribution from counting the possible arrangements of successes and failures.

Example

100 emails are sent.

Probability of a user clicking:

$$p = 0.1$$

Let X= number of clicks.

Then:

$$X \sim Binomial(100, 0.1)$$

Question:

What is the probability of exactly 10 clicks?

$$P(X = 10) = \binom{100}{10}(0.1)^{10}(0.9)^{90}$$

Python

```
from scipy.stats import binom


prob = binom.pmf(k=10, n=100, p=0.1)


print(prob)
```

Interview question

Bernoulli vs Binomial?

Bernoulli models one binary trial. Binomial models the number of successes across a fixed number $n$of independent Bernoulli trials.

---

9. Geometric Distribution

P1

Geometric distribution answers:

How many trials do I need before the first success?

The source defines it using repeated coin tosses until the first head.

$$P(X = k) = (1 - p)^{k-1}p$$

Diagram

Trial 1 → Failure

Trial 2 → Failure

Trial 3 → Failure

Trial 4 → SUCCESS

|

▼

X = 4

Example

A server request has probability:

$$p = 0.2$$

of succeeding.

Probability first success occurs on request 4:

$$P(X = 4) = (0.8)^3(0.2)$$

Python

```
from scipy.stats import geom


prob = geom.pmf(4, 0.2)


print(prob)
```

---

10. Uniform Distribution

P1

Uniform means outcomes within the specified range are equally likely.

Discrete example:

Dice

1 2 3 4 5 6

| | | | | |

1/6 each

The source uses a six-sided die as the basic example.

Continuous example:

$$X \sim Uniform(a, b)$$

Every interval's probability is proportional to its length.

ML example

Suppose a simulation randomly chooses:

$$X \in [0, 1]$$

Then:

$$P(0.2 < X < 0.5) = 0.3$$

Python

```
from scipy.stats import uniform


prob = uniform.cdf(0.5) - uniform.cdf(0.2)


print(prob)
```

---

11. PDF — Probability Density Function

P0

This is one of the most common interview traps.

A PDF is **NOT itself a probability**.

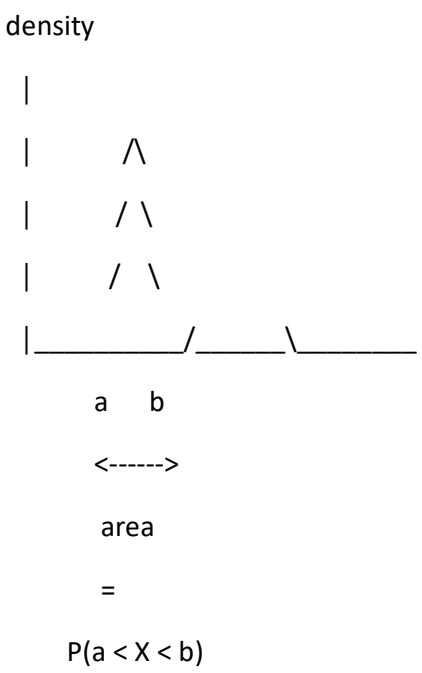
The source explicitly emphasizes this distinction.

For a continuous random variable:

$$P(a < X < b) = \int_a^b f_X(x)\, dx$$

Think:

PDF



Critical interview point

For a continuous variable:

$$P(X = x) = 0$$

for an individual point.

Probability comes from **area under the curve**, not the height of the PDF.

The source explicitly explains why individual points can have probability zero in continuous distributions.

Common interview question

Q: Is PDF a probability?

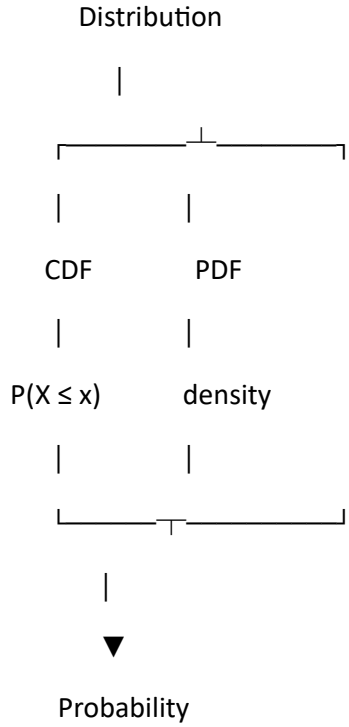
Answer:

No. A PDF is a density. The probability of an interval is the integral of the PDF over that interval.

---

12. CDF vs PDF

P0



For a continuous distribution:

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$$

and, where differentiable,

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Interview answer

CDF gives cumulative probability. PDF gives probability density. For a continuous variable, probabilities over intervals are obtained by integrating the PDF, while the CDF is the accumulated integral of the PDF.

---

13. Normal Distribution

P0 — VERY IMPORTANT

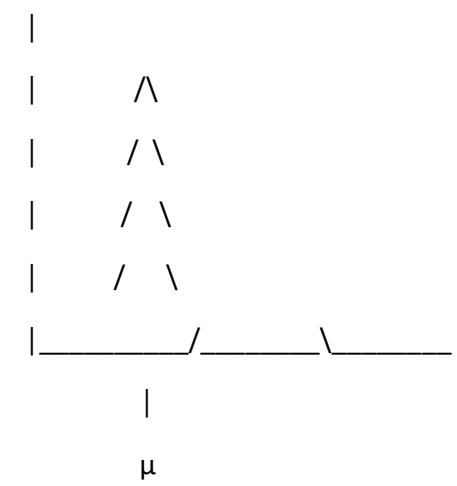
The normal distribution is one of the most important distributions in statistics and ML; the source explicitly highlights its importance.

$$X \sim N(\mu, \sigma^2)$$

where:

- μ= mean
- σ^2= variance
- σ= standard deviation

Shape



Why important?

Many naturally occurring measurements are approximately normal, and the normal distribution appears extensively in statistical theory.

Python

```
from scipy.stats import norm


mu = 100

sigma = 15


prob = norm.cdf(120, loc=mu, scale=sigma)


print(prob)
```

This calculates:

$$P(X \leq 120)$$

---

14. Independence

P0 — VERY IMPORTANT

Two random variables $X$and $Y$are independent if:

$$P(X, \in AY \in B) = P(X \in A)P(Y \in B)$$

The source gives this definition directly.

Intuition

X happens

|

| no influence

▼

Y happens

Knowing Xtells us nothing about Y.

Example

Two independent coin tosses:

Toss 1 ————> X

\

\

independent

/

/

Toss 2 ————> Y

ML example

Suppose:

- X= result of today's coin toss
- Y= result of tomorrow's independent coin toss

Then:

$$P(X, = HY = H) = P(X = H)P(Y = H)$$

Important warning

Correlation = 0 does NOT generally imply independence.

But independence implies zero covariance when the relevant moments exist.

That distinction is highly interview-worthy.

---

15. i.i.d.

P0

i.i.d. = independent and identically distributed

X_1 X_2 X_3 X_4

same distribution

+ independent

Mathematically:

$$X_1, X_2, \dots, X_n P$$

ML example

Training samples are often modeled as:

$$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$$

drawn independently from the same underlying data-generating distribution.

Interview answer

i.i.d. means every observation follows the same distribution and the observations are independent of one another.

The source explicitly notes that sequences of random variables are often assumed to be i.i.d.

---

16. Law of Total Probability

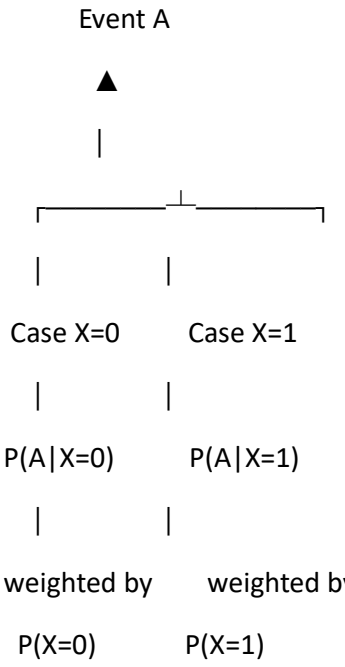
P0 — VERY IMPORTANT

If different cases partition the possibilities:

$$P(A) = \sum_{k} P\,(A \mid X = x_k) P(X = x_k)$$

The source presents this exact random-variable formulation.

Diagram



Example

Suppose:

Traffic source:

Organic 70%

Paid 30%

Conversion rates:

Organic → 5%

Paid → 10%

Overall conversion:

$$P(C) = P(C \mid Organic) P(Organic) + P(C \mid Paid) P(Paid)$$

$$= 0.05(0.7) + 0.10(0.3)$$
$$= 0.065$$

So overall conversion probability is 6.5%.

Interview insight

This formula is essentially:

weighted average of conditional probabilities

---

17. Convolution

P1 — Important

When adding independent random variables, their distributions combine through **convolution**.

The source explicitly derives the distribution of $X + Y$ using convolution.

For discrete variables:

$$P(X + Y = k) = \sum_{l} P\,(X = k - l)P(Y = l)$$

or:

$$p_{X+Y} = p_X * p_Y$$

Dice example

Dice 1 Dice 2

X Y

| |

└──────┬──────┘

▼

X + Y

|

▼

New distribution

For two dice:

Sum

2 3 4 5 6 7 8 9 10 11 12



Python

```
import numpy as np


dice = np.array([0, 1/6, 1/6, 1/6, 1/6, 1/6, 1/6])


sum_distribution = np.convolve(dice, dice)


print(sum_distribution)
```

The source itself uses np.convolve for this exact dice example.

---

18. Binomial as Sum of Bernoulli Variables

P0 — Great interview connection

This is more important than memorizing the binomial formula.

Suppose:

$$X_i \sim Bernoulli(p)$$

and the trials are independent.

Then:

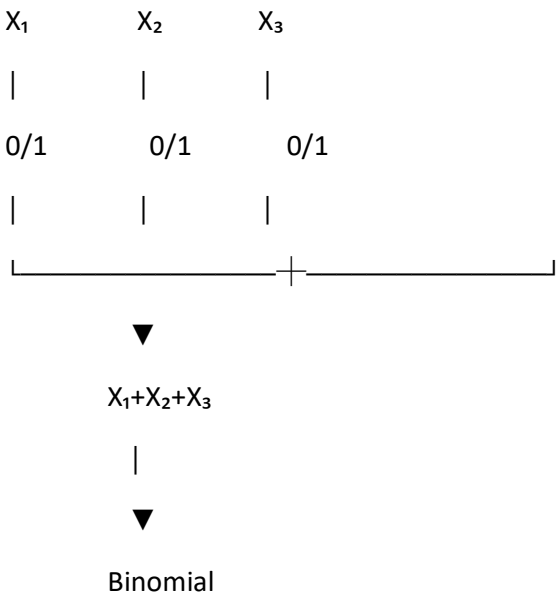
$$X = X_1 + X_2 + \cdots + X_n$$

has:

$$X \sim Binomial(n, p)$$

Diagram

Bernoulli Bernoulli Bernoulli



The source derives the binomial distribution precisely by summing independent Bernoulli random variables.

This is an excellent interview answer because it demonstrates **understanding rather than memorization**.

---

19. Mixed Distribution

P2

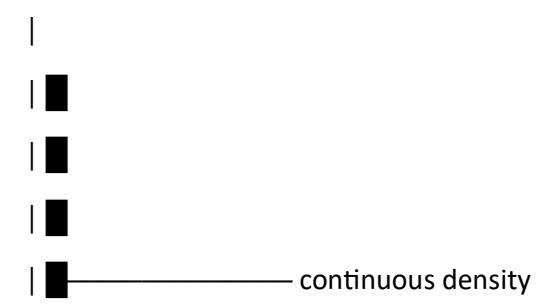
Not every random variable is purely discrete or purely continuous.

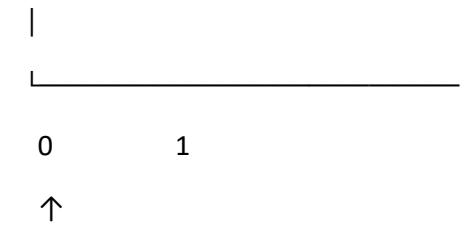
The source gives an example where:

- with probability 1/2, $X = 0$
- otherwise $X$is uniformly distributed between 0 and 1

This produces a **mixed distribution**.

Diagram:





This is useful to know conceptually, but don't spend much interview-preparation time here unless your role is statistics-heavy.

---

20. Distribution Cheat Sheet

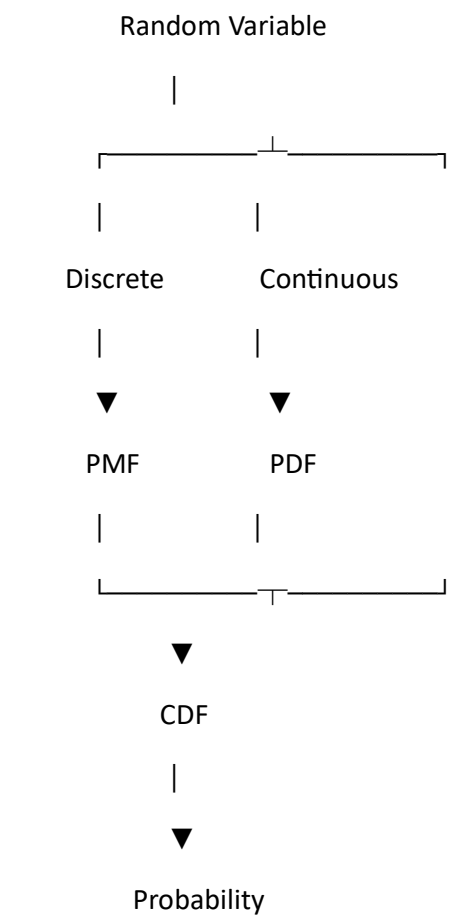
Concept	Meaning	Example	Priority
Random variable	Maps outcome → value	customer spending	P0
PMF	$P(X = x)$	number of clicks	P0
CDF	$P(X \leq x)$	latency below threshold	P0
PDF	Probability density	height	P0
Bernoulli	One binary trial	fraud/not fraud	P0
Binomial	Successes in n trials	clicks in 100 emails	P0
Normal	Bell-shaped continuous distribution	measurement noise	P0
Independence	One variable doesn't affect another	independent tosses	P0
i.i.d.	independent + same distribution	training samples	P0
Total probability	Weighted conditional probabilities	conversion rate	P0
Geometric	Trials until first success	retries until success	P1
Uniform	Equal density/rates over interval	random simulation	P1
Convolution	Distribution of sum	sum of dice	P1
Mixed distribution	Discrete + continuous	point mass + interval	P2

---

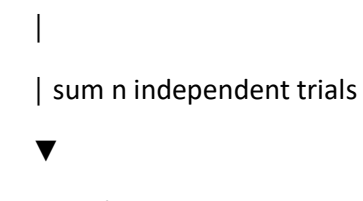
21. The Most Important Interview Relationships

Don't memorize these as isolated formulas.

Understand this chain:



And:



And:



|

▼

Distribution of X+Y

And:

Conditional probabilities

|

▼

Law of Total Probability

|

▼

Overall probability

---

22. Implementation — 80/20 Python

You should be able to implement the basic ideas **without SciPy**.

Bernoulli

```
import random


def bernoulli(p):
    return 1 if random.random() < p else 0


samples = [bernoulli(0.7) for _ in range(10)]


print(samples)
```

Empirical PMF

Suppose we have observations:

```
from collections import Counter
```

```
samples = [1, 0, 1, 1, 0, 1, 0, 0, 1, 1]


counts = Counter(samples)


n = len(samples)


pmf = {
    value: count / n
    for value, count in counts.items()
}


print(pmf)
```

Example output:

```
{
    1: 0.6,
    0: 0.4
}
```

This demonstrates an important ML idea:

Samples

  |

  ▼

Count occurrences

  |

  ▼

Divide by total observations

  |

  ▼

Empirical distribution

---

23. Implement CDF Yourself

```
def empirical_cdf(samples, x):
    count = sum(value <= x for value in samples)
    return count / len(samples)


samples = [1, 2, 2, 3, 4, 5]

print(empirical_cdf(samples, 3))
```

For:

1, 2, 2, 3, 4, 5

we have:

$$P(X \leq 3) = \frac{4}{6} = \frac{2}{3}$$

---

24. Implement Binomial Probability

```
import math

def binomial_pmf(n, k, p):
    combinations = math.comb(n, k)

    return combinations * (p ** k) * ((1 - p) ** (n - k))


print(binomial_pmf(10, 3, 0.5))
```

This is the formula:

$$\binom{n}{k} p^k (1-p)^{n-k}$$

---

25. Implement Discrete Convolution

```
def convolution(a, b):
    result = [0] * (len(a) + len(b) - 1)

    for i, x in enumerate(a):
        for j, y in enumerate(b):
            result[i + j] += x * y

    return result
```

Example:

```
dice = [1/6] * 6

result = convolution(dice, dice)

print(result)
```

This is the core operation behind combining discrete independent distributions.

---

26. Interview Questions You Should Practice

P0 Questions

Q1. What is a random variable?

Answer:

A random variable is a function that maps outcomes of a random experiment to measurable values.

Q2. What is PMF?

PMF gives the probability of each exact value of a discrete random variable:

$$P(X = x)$$

Q3. What is CDF?

CDF gives the probability that a random variable is less than or equal to a value:

$$F_X(x) = P(X \leq x)$$

Q4. PMF vs PDF?

PMF is used for discrete random variables and directly gives point probabilities. PDF is used for continuous variables and gives probability density; probabilities are obtained by integrating the PDF over an interval.

Q5. Can PDF be greater than 1?

Yes.

This is a classic interview trap.

A density can exceed 1 as long as its total area is 1.

For example:

$$f(x) = 2,\ 0 < x < 0.5$$

has:

$$\int_0^{0.5} 2\,dx = 1$$

So it is a valid PDF.

Q6. Is $P(X = x) = 0$possible even though x is a valid value?

Yes, for continuous random variables.

The probability is represented by area over intervals.

Q7. Bernoulli vs Binomial?

Bernoulli represents one binary trial. Binomial represents the number of successes across $n$independent Bernoulli trials.

Q8. What does independence mean?

Knowing the value of one variable does not change the probability distribution of the other.

Mathematically:

$$P(X, Y) = P(X)P(Y)$$

for the relevant events.

Q9. What does i.i.d. mean?

Independent and identically distributed.

Q10. Why is the normal distribution important?

It is fundamental in statistical modeling and theory and appears in many practical measurement/noise settings and statistical approximations.

The source specifically identifies it as the most important distribution in statistics.

---

27. High-Value Interview Traps

Memorize these **conceptual distinctions**, not just formulas.

Trap 1

❌ "PDF gives probability."

✅ "PDF gives density; area gives probability."

Trap 2

❌ "A random variable is a random number."

✅ "A random variable is a function mapping outcomes to values."

Trap 3

❌ "Binomial means repeated random numbers."

✅ "Binomial counts successes across a fixed number of independent Bernoulli trials."

Trap 4

❌ "CDF gives probability exactly at x."

✅

$$F(x) = P(X \leq x)$$

Trap 5

❌ "Continuous variables cannot have probability."

✅ They have probability over **ranges/intervals**.

Trap 6

❌ "Zero correlation means independence."

✅ In general, **zero correlation does not imply independence**.

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28. What You Should Actually Memorize

For an interview, don't attempt to memorize the entire chapter.

Memorize this:

RANDOM VARIABLES

```
    |
    ├── Discrete
    |     |
    |     └── PMF
    |
    └── Continuous
          |
          └── PDF
```

CDF:

$F(x) = P(X \leq x)$

Discrete:

$P(X = x) = PMF$

Continuous:

$P(a < X < b) = \int f(x)\, dx$

BERNOULLI:

one binary trial

BINOMIAL:

n independent Bernoulli trials

number of successes

GEOMETRIC:

trials until first success

NORMAL:

$N(\mu, \sigma^2)$

INDEPENDENCE:

$P(X,Y) = P(X)P(Y)$

i.i.d.:

independent + identical distribution

TOTAL PROBABILITY:

$P(A) = \Sigma P(A|Xi)P(Xi)$

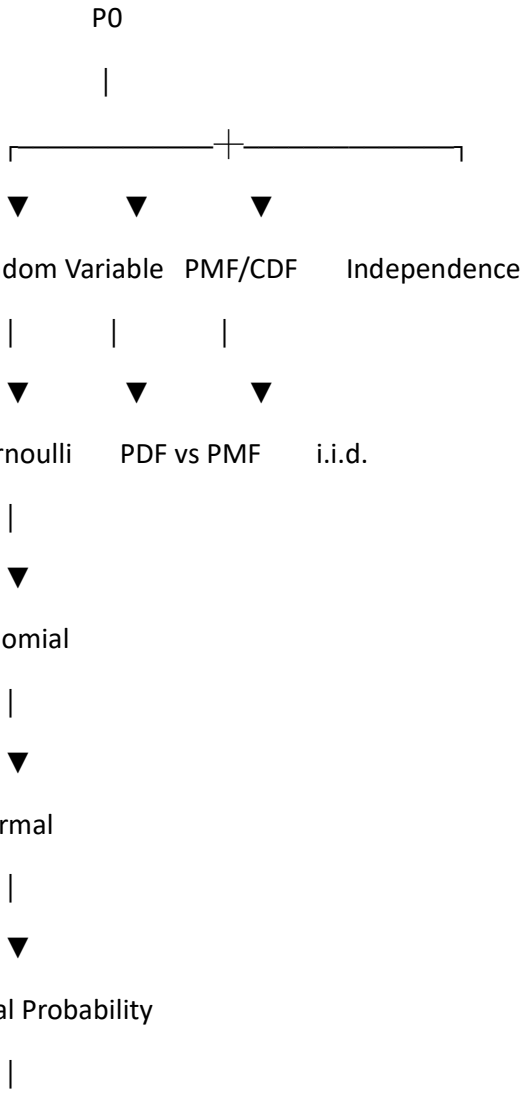
CONVOLUTION:

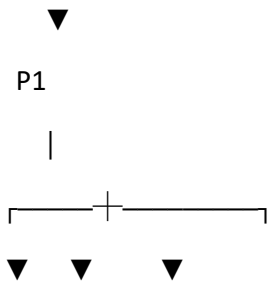
distribution of X + Y

The source's central progression is exactly toward representing random variables through distributions because distributions are much easier to manipulate computationally and mathematically.

Your 80/20 priority

Study in this order:





For your next ML/AI interview, the highest-value skill is being able to take a real problem and identify the random variable, its possible values, appropriate distribution, and what probability question you are actually trying to answer.